

Raghav Gupta

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About

Second-year IT engineering student focused on practical AI: fine-tuning LLMs, building RAG + ASR pipelines, and multimodal emotion recognition. Fast learner, shipping usable systems early.

Education

Bachelor of Engineering in Information Technology

2023 – 2027

University Institute of Engineering and Technology, Panjab University

- Reliance Foundation Undergraduate Scholar

Experience

AI/ML Developer | Design Innovation Centre (DIC)

Jan 2024 – Present

- Developed computer vision pipelines using OpenCV, Ffmpeg, imgaug, etc. for keyframe extraction and video augmentation, supporting model training and making it more efficient.
- Built and optimized facial emotion recognition systems using Action Unit detection with pretrained DL models, improving inference accuracy.
- Fine-tuned and deployed domain-specific Large Language Models (LLMs) using HuggingFace Transformers and PyTorch for document understanding and response generation.
- Researched and prototyped multimodal deep learning models (audio + vision) for enhanced emotion recognition accuracy.

AI Intern | Lafleurtech

Jun 2024 – Aug 2024

- Developed a local ASR transcription pipeline integrating Whisper and NVIDIA Parakeet models with noise reduction, speaker diarization, and batching for efficient processing.

Open Source Backend Intern | AISOC / PClub UIET

July 2024 – Aug 2024

- Engineered the model-hosting layer of PU-API, an internal API platform consolidating deployment of DIC and university AI services (LLMs, CV, ASR) with containerized inference endpoints.

ML Developer | Hackathons – RaG Systems

Feb 2024

- Designed and implemented a RaG system using FAISS and LLMs to extract insights from business documents, enabling automated decision support for industrial clients.

Open Source Contributor | GitHub Projects

2020 – Present

- Active GitHub contributor with 500+ commits in the past years across ML tools, automation scripts, and student-focused repositories.

Technical Skills

Programming: Python, C, C++, JavaScript, TypeScript, Rust, Haskell, SQL

Libraries/Frameworks: PyTorch, TensorFlow, HuggingFace, OpenCV, Whisper, NVIDIA Parakeet, Scikit-learn, FAISS

Dev / Ops: Git, Docker, GitHub Actions, FastAPI, REST, VS Code, WSL, Jupyter, LaTeX

Domain: ASR, Multimodal Emotion Recognition, LLM Fine-tuning, RAG, Video Processing, Model Hosting

Additional Information

Research Interests: Speech Processing, Audio Analysis, Deep Learning, Interpretability, Self-evolving Agents, Just-in-Time Generation

Soft Skills: Teamwork, Problem-solving, Critical Thinking, Communication